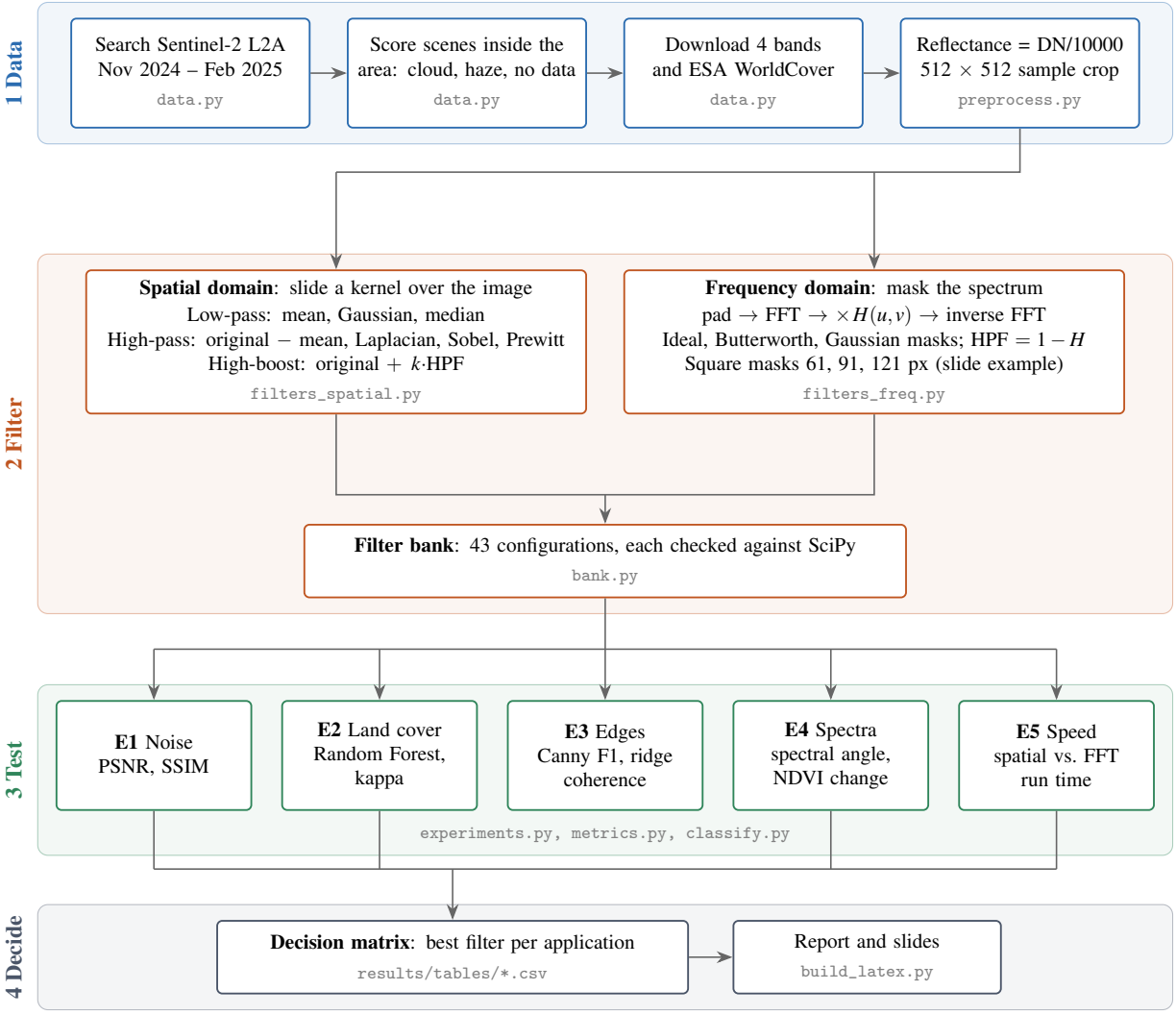

1 Data
Search Sentinel-2 L2A
Nov 2024 – Feb 2025
data.py
Score scenes inside the
area: cloud, haze, no data
data.py
Download 4 bands
and ESA WorldCover
data.py
Reflectance = DN/10000
512 × 512 sample crop
preprocess.py
2 Filter
Spatial domain: slide a kernel over the image
Low-pass: mean, Gaussian, median
High-pass: original − mean, Laplacian, Sobel, Prewitt
High-boost: original + k·HPF
filters_spatial.py
Frequency domain: mask the spectrum
pad → FFT → ×H(u,v) → inverse FFT
Ideal, Butterworth, Gaussian masks; HPF = 1 − H
Square masks 61, 91, 121 px (slide example)
filters_freq.py
Filter bank: 43 configurations, each checked against SciPy
bank.py
3 Test
E1 Noise
PSNR, SSIM
E2 Land cover
Random Forest,
kappa
E3 Edges
Canny F1, ridge
coherence
E4 Spectra
spectral angle,
NDVI change
E5 Speed
spatial vs. FFT
run time
experiments.py, metrics.py, classify.py
4 Decide
Decision matrix: best filter per application
results/tables/*.csv
Report and slides
build_latex.py